Text Processing Exercise

A support team receives short incident notes. Each note begins with a ticket identifier that should be extracted as a structured field, not treated as a text feature.

doc_id	raw_note
D1	TKT-1001 vpn not working today
D2	TKT-1002 vpn not working now
D3	TKT-1003 vpn not responding today
D4	TKT-1004 printer working fine now
D5	TKT-1005 printer not working

Exercises

Unless stated otherwise, use base-10 logarithms.

1. **Use the regex `\bTKT-\d{4}\b` to extract the ticket id from each raw_note. For every document, show ticket_id, cleaned feature text after removing the ticket id, the token list after lowercasing and splitting on whitespace, and the token length (L_d).**

doc_id	ticket_id	cleaned feature text	tokens
D1	TKT-1001	vpn not working today	vpn, not, working, today
D2	TKT-1002	vpn not working now	vpn, not, working, now
D3	TKT-1003	vpn not responding today	vpn, not, responding, today
D4	TKT-1004	printer working fine now	printer, working, fine, now
D5	TKT-1005	printer not working	printer, not, working

The document lengths are:

doc_id	token length L_d
D1	4
D2	4
D3	4
D4	4
D5	3

2. **Using the tokens from Exercise 1, calculate document frequency for every unique unigram. Show a table with `term`, documents containing the term, `df` and IDF.**

term	documents	df	idf calculation	idf
vpn	D1, D2, D3	3	log10(6/4)	0.176
not	D1, D2, D3, D5	4	log10(6/5)	0.079
working	D1, D2, D4, D5	4	log10(6/5)	0.079
today	D1, D3	2	log10(6/3)	0.301
now	D2, D4	2	log10(6/3)	0.301
responding	D3	1	log10(6/2)	0.477
printer	D4, D5	2	log10(6/3)	0.301
fine	D4	1	log10(6/2)	0.477

3. **Calculate the unigram TF-IDF score for every term that appears in each document. Show the full score table and identify the highest unigram signals.**

Every term appears once in its document, so the local term count is 1. For D1:

$$\mathrm{tf}(\text{vpn}, D1) = \frac{1}{4} = 0.25$$

$$\mathrm{tfidf}(\text{vpn}, D1) = 0.25 \cdot 0.176 = 0.044$$

$$\mathrm{tfidf}(\text{today}, D1) = 0.25 \cdot 0.301 = 0.075$$

For D3:

$$\mathrm{tfidf}(\text{responding}, D3) = 0.25 \cdot 0.477 = 0.119$$

The complete selected unigram score table is:

doc_id	term	tf	idf	tf-idf
D1	vpn	0.250	0.176	0.044
D1	not	0.250	0.079	0.020
D1	working	0.250	0.079	0.020
D1	today	0.250	0.301	0.075
D2	vpn	0.250	0.176	0.044
D2	not	0.250	0.079	0.020
D2	working	0.250	0.079	0.020
D2	now	0.250	0.301	0.075
D3	vpn	0.250	0.176	0.044
D3	not	0.250	0.079	0.020
D3	responding	0.250	0.477	0.119
D3	today	0.250	0.301	0.075

doc_id	term	tf	idf	tf-idf
D4	printer	0.250	0.301	0.075
D4	working	0.250	0.079	0.020
D4	fine	0.250	0.477	0.119
D4	now	0.250	0.301	0.075
D5	printer	0.333	0.301	0.100
D5	not	0.333	0.079	0.026
D5	working	0.333	0.079	0.026

4. **For each cleaned document, generate all contiguous two-token windows. Show the bigram list for each document and calculate the bigram denominator.**

For bigrams, n=2:

$$m_2(d) = L_d - 1$$

doc_id	tokens	bigram windows m_2(d)	bigrams
D1	vpn not working today	3	vpn not, not working, working today
D2	vpn not working now	3	vpn not, not working, working now
D3	vpn not responding today	3	vpn not, not responding, responding today
D4	printer working fine now	3	printer working, working fine, fine now
D5	printer not working	2	printer not, not working

For a bigram g:

$$\mathrm{tf}_2(g, d) = \frac{f_{g,d}}{m_2(d)}$$

The important denominator is the number of bigram windows, not the number of tokens.

5. **Using the bigrams from Exercise 4, calculate document frequency and IDF for every unique bigram. Then calculate bigram TF-IDF for every bigram that appears in each document.**

bigram	documents	df	idf
vpn not	D1, D2, D3	3	0.176
not working	D1, D2, D5	3	0.176
working today	D1	1	0.477

bigram	documents	df	idf
working now	D2	1	0.477
not responding	D3	1	0.477
responding today	D3	1	0.477
printer working	D4	1	0.477
working fine	D4	1	0.477
fine now	D4	1	0.477
printer not	D5	1	0.477

For D1:

$$\mathrm{tf}_2(\text{not working}, D1) = \frac{1}{3} = 0.333$$

$$\mathrm{tfidf}_2(\text{not working}, D1) = 0.333 \cdot 0.176 = 0.059$$

$$\mathrm{tfidf}_2(\text{working today}, D1) = 0.333 \cdot 0.477 = 0.159$$

For D5:

$$\mathrm{tf}_2(\text{not working}, D5) = \frac{1}{2} = 0.500$$

$$\mathrm{tfidf}_2(\text{not working}, D5) = 0.500 \cdot 0.176 = 0.088$$

The complete bigram score table is:

doc_id	bigram	tf	idf	bigram tf-idf
D1	vpn not	0.333	0.176	0.059
D1	not working	0.333	0.176	0.059
D1	working today	0.333	0.477	0.159
D2	vpn not	0.333	0.176	0.059
D2	not working	0.333	0.176	0.059
D2	working now	0.333	0.477	0.159
D3	vpn not	0.333	0.176	0.059
D3	not responding	0.333	0.477	0.159
D3	responding today	0.333	0.477	0.159
D4	printer working	0.333	0.477	0.159
D4	working fine	0.333	0.477	0.159
D4	fine now	0.333	0.477	0.159
D5	printer not	0.500	0.477	0.239
D5	not working	0.500	0.176	0.088

6. Add `<s>` at the start and `</s>` at the end of each cleaned sentence. Count every transition from history `h` to next token `w`, then show `c(h,w)` and the total count `c(h)`.

doc_id	boundary sequence
D1	<s> vpn not working today </s>
D2	<s> vpn not working now </s>
D3	<s> vpn not responding today </s>
D4	<s> printer working fine now </s>
D5	<s> printer not working </s>

The bigram model uses a first-order Markov assumption:

$$P(w_{1:T}) \approx \prod_{t=1}^{T+1} P\left(w_t \mid w_{t-1}\right), \qquad w_0 = \langle s \rangle, \quad w_{T+1} = \langle /s \rangle$$

The count-based maximum-likelihood estimate is:

$$\hat{P}(w \mid h) = \frac{c(h,w)}{c(h)}$$

history h	next-token counts	total c(h)
<s>	vpn: 3, printer: 2	5
vpn	not: 3	3
not	working: 3, responding: 1	4
working	today: 1, now: 1, fine: 1, </s>: 1	4
today	</s>: 2	2
now	</s>: 2	2
printer	working: 1, not: 1	2
responding	today: 1	1
fine	now: 1	1

Selected probabilities are:

$$\hat{P}(\text{vpn} \mid \langle s \rangle) = \frac{3}{5} = 0.600$$

$$\hat{P}(\text{not} \mid \text{vpn}) = \frac{3}{3} = 1.000$$

$$\hat{P}(\text{working} \mid \text{not}) = \frac{3}{4} = 0.750$$

$$\hat{P}(\text{today} \mid \text{working}) = \frac{1}{4} = 0.250$$

$$\hat{P}(\langle/s\rangle \mid \text{today}) = \frac{2}{2} = 1.000$$

7. Calculate the probability of the sentence `vpn not working today </s>` given `<s>`.

Compute the probability of the sentence `vpn not working today`:

$$P(\text{vpn not working today } \langle/s\rangle \mid \langle s\rangle)$$

$$= \hat{P}(\text{vpn} \mid \langle s\rangle)\hat{P}(\text{not} \mid \text{vpn})\hat{P}(\text{working} \mid \text{not})\hat{P}(\text{today} \mid \text{working})\hat{P}(\langle/s\rangle \mid \text{today})$$

$$= \frac{3}{5} \cdot \frac{3}{3} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{2}{2}$$

$$= \frac{9}{80} = 0.1125$$

8. For `printer not responding </s>`, identify the unseen transition that makes the unsmoothed probability zero. Then use add-one smoothing with |V|=9 to calculate the smoothed probability.

The sentence `printer not responding` contains the unseen transition `responding -> </s>`.

$$\hat{P}(\langle/s\rangle \mid \text{responding}) = \frac{0}{1} = 0$$

Therefore the whole sentence receives probability zero under the unsmoothed model.

One simple smoothing option is add-one smoothing:

$$\hat{P}_{+1}(w \mid h) = \frac{c(h, w) + 1}{c(h) + |V|}$$

If the next-token vocabulary is:

$$V = \{\text{vpn,printer,not,working,responding,today,now,fine,} \langle/s\rangle\}$$

then |V|=9. The smoothed probability of `printer not responding </s>` is:

$$P_{+1}(\text{printer not responding } \langle/s\rangle \mid \langle s\rangle)$$

$$= \frac{2+1}{5+9} \cdot \frac{1+1}{2+9} \cdot \frac{1+1}{4+9} \cdot \frac{0+1}{1+9}$$

$$= \frac{3}{14} \cdot \frac{2}{11} \cdot \frac{2}{13} \cdot \frac{1}{10}$$

$$\approx 0.0006$$

9. Use Q=`[0.80,0.10,0.60]`, D1=`[0.90,0.10,0.50]`, and D4=`[0.10,0.90,0.20]`. Calculate cosine similarity for Q against D1 and D4 by showing the dot product, vector norms, and final similarity.

item	text	vector
query Q	vpn outage	[0.80, 0.10, 0.60]
ticket D1	vpn not working today	[0.90, 0.10, 0.50]
ticket D4	printer working fine now	[0.10, 0.90, 0.20]

Cosine similarity is:

$$\cos(a,b) = \frac{a \cdot b}{\| a \| \| b \|}$$

<u>Query compared with D1</u>

Dot product:

$$Q \cdot D1 = (0.80)(0.90) + (0.10)(0.10) + (0.60)(0.50) = 1.030$$

Vector norms:

$$\| Q \| = \sqrt{0.80^2 + 0.10^2 + 0.60^2} = \sqrt{1.01} = 1.005$$

$$\| D1 \| = \sqrt{0.90^2 + 0.10^2 + 0.50^2} = \sqrt{1.07} = 1.034$$

Cosine similarity:

$$\cos(Q, D1) = \frac{1.030}{(1.005)(1.034)} \approx 0.991$$

<u>Query compared with D4</u>

Dot product:

$$Q \cdot D4 = (0.80)(0.10) + (0.10)(0.90) + (0.60)(0.20) = 0.290$$

Vector norm:

$$\| D4 \| = \sqrt{0.10^2 + 0.90^2 + 0.20^2} = \sqrt{0.86} = 0.927$$

Cosine similarity:

$$\cos(Q, D4) = \frac{0.290}{(1.005)(0.927)} \approx 0.311$$